

Lecture Example 2 - Tabular Simplex

Tableau Simplex (Big-M) solution report · generated 2026-09-27 13:54

Lecture slides 15-57. All $\leq$ constraints, slack variables give the starting basis. Three iterations to the optimum $Z = 70$.

Model

Maximize $Z = 20X_1 + 10X_2$
subject to
[C1] $X_1 - X_2 \leq 1$
[C2] $3X_1 + X_2 \leq 7$
 $X_1, X_2 \geq 0$

RESULT

OPTIMAL

$Z^* = 70$

$X_1 = 0$

$X_2 = 7$

iterations: 3 · rule: dantzig

Findings

- Optimal: row 0 has no negative coefficients. $Z^* = 70$

Contents: standard form · geometry · every tableau with entering/leaving analysis · decision diagram · summary dashboard · independent verification

1. Converting to the standard LP form

1. C1 (Type I, $\leq$): add slack $X3 \geq 0$ (unused resource, cost 0) $\rightarrow$ $X3$ is the starting basic variable of this row.
2. C2 (Type I, $\leq$): add slack $X4 \geq 0$ (unused resource, cost 0) $\rightarrow$ $X4$ is the starting basic variable of this row.

Maximize $Z = 20X1 + 10X2$

subject to

$$X1 - X2 + X3 = 1$$

$$3X1 + X2 + X4 = 7$$

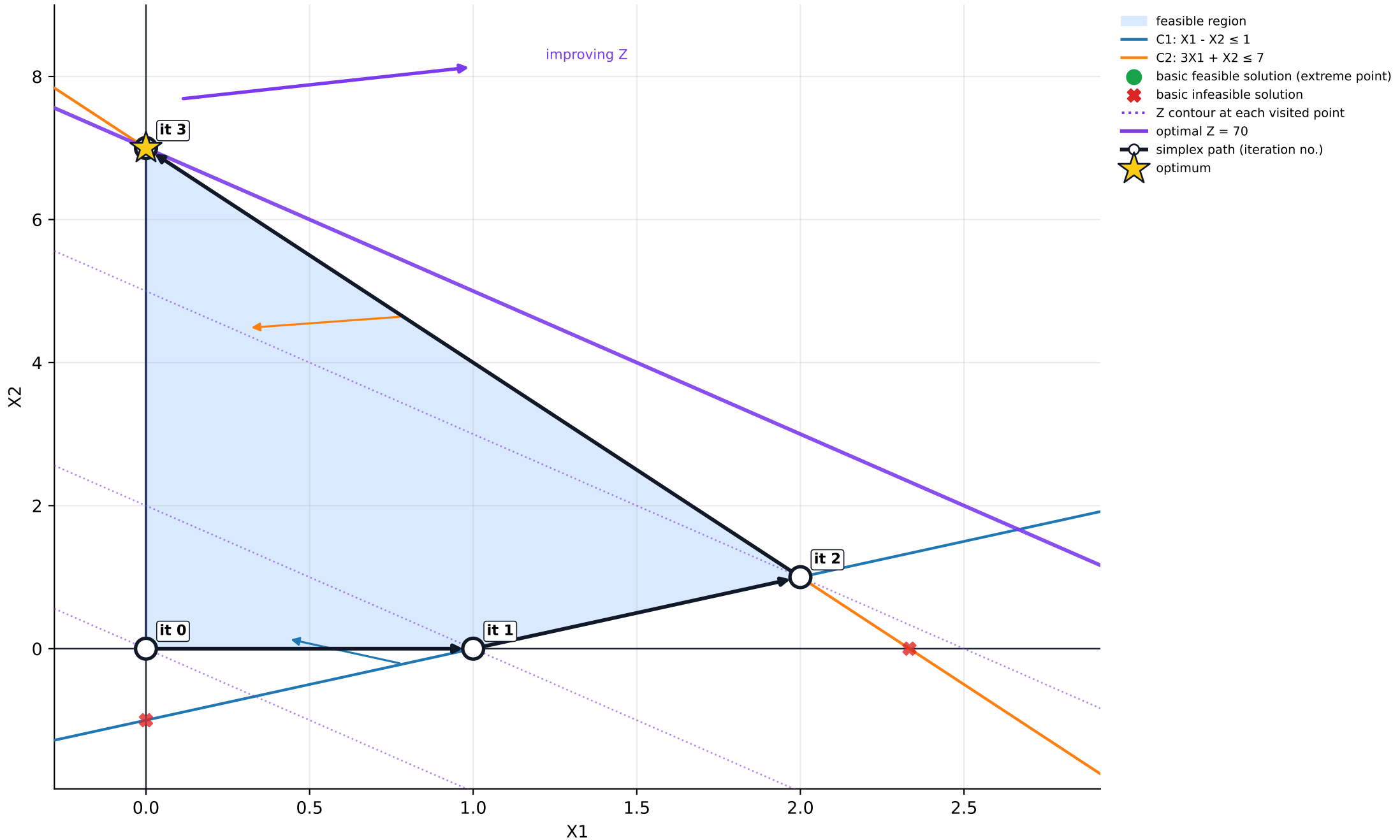
$$X1, X2, X3, X4 \geq 0$$

Standard form - coefficient matrix

	X1	X2	X3	X4		RHS
max Z	20	10	0	0		
C1	1	-1	1	0	=	1
C2	3	1	0	1	=	7

2. Geometric view: basic solutions, extreme points and the simplex path

Geometry of the simplex path - optimal



3. All basic solutions (m variables basic, the rest 0). Green = feasible = extreme point; red = infeasible; ★ = optimal

Basic variables	X1	X2	Feasible?	Z	X3	X4	Optimal?
X1, X2	2	1	yes	50	0	0	
X1, X3	2.3333	0	no	46.6667	-1.3333	0	
X1, X4	1	0	yes	20	0	4	
X2, X3	0	7	yes	70	8	0	★
X2, X4	0	-1	no	-10	0	8	
X3, X4	0	0	yes	0	1	7	

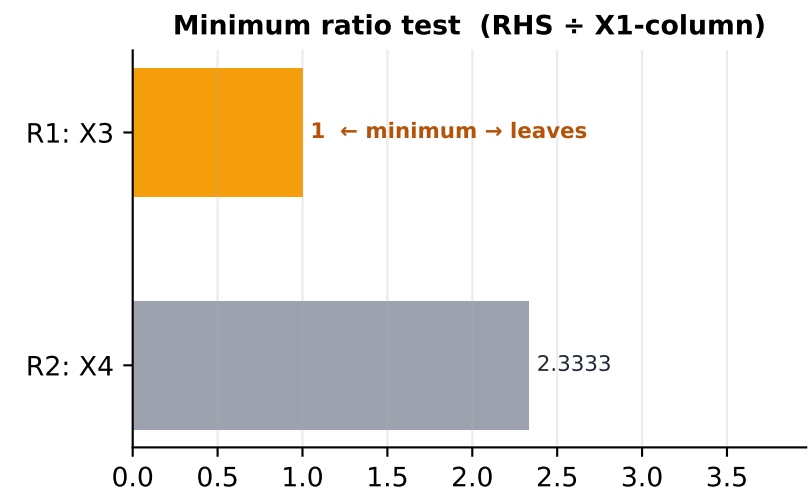
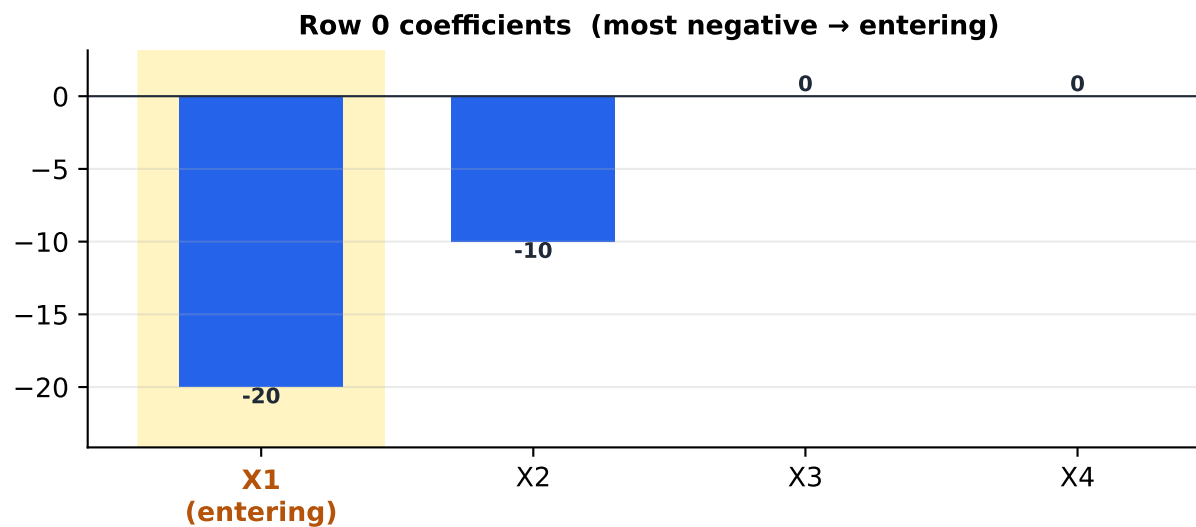
Initial tableau

Basic	Eq.	Z	X1	X2	X3	X4	RHS
Z	0	1	-20	-10	0	0	0
X3	1	0	1	-1	1	0	1
X4	2	0	3	1	0	1	7

Initial tableau. Basic variables: $X_3 = 1$, $X_4 = 7$. Non-basic (set to 0): X_1 , X_2 . The values of the basic variables are read directly from the RHS column.

Iteration 1: entering & leaving variables

Basic	Eq.	Z	X1	X2	X3	X4	RHS	Ratio
Z	0	1	-20	-10	0	0	0	
X3	1	0	1	-1	1	0	1	1
X4	2	0	3	1	0	1	7	2.3333



Iteration 1 - selecting variables.

- Is the current solution optimal? No - row 0 has at least one negative coefficient.
- Entering basic variable: X1 - it has the largest (in absolute value) negative coefficient in row 0 (-20).
- Minimum ratio test (RHS ÷ positive entries of the X1-column): R1: 1, R2: 2.3333.
- Leaving basic variable: X3 (row 1, smallest ratio). Pivot number = 1.

Iteration 1: pivoting (Gauss-Jordan row operations)

(a) pivot row scaled: Pivot number is already 1 - scaling not needed (normally divide the pivot row by the pivot number)

Basic	Eq.	Z	X1	X2	X3	X4	RHS	Ratio
Z	0	1	-20	-10	0	0	0	
X3	1	0	1	-1	1	0	1	1
X4	2	0	3	1	0	1	7	2.3333

(b) X1 replaced X3 in the basis. Z = 20

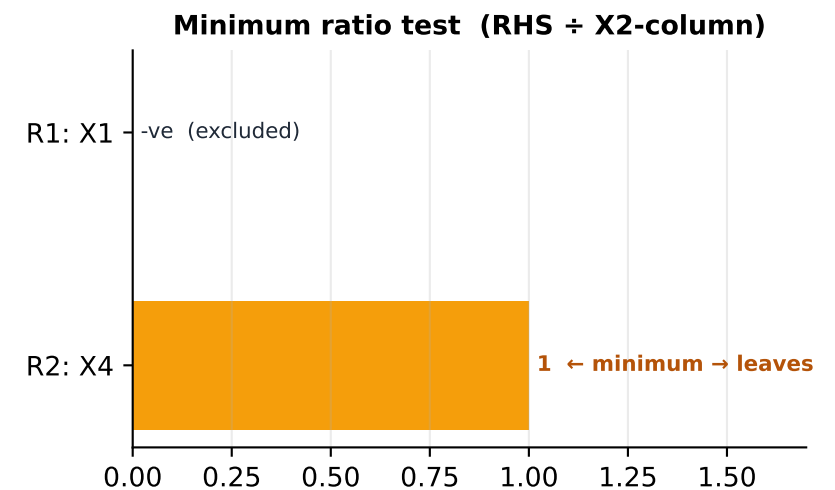
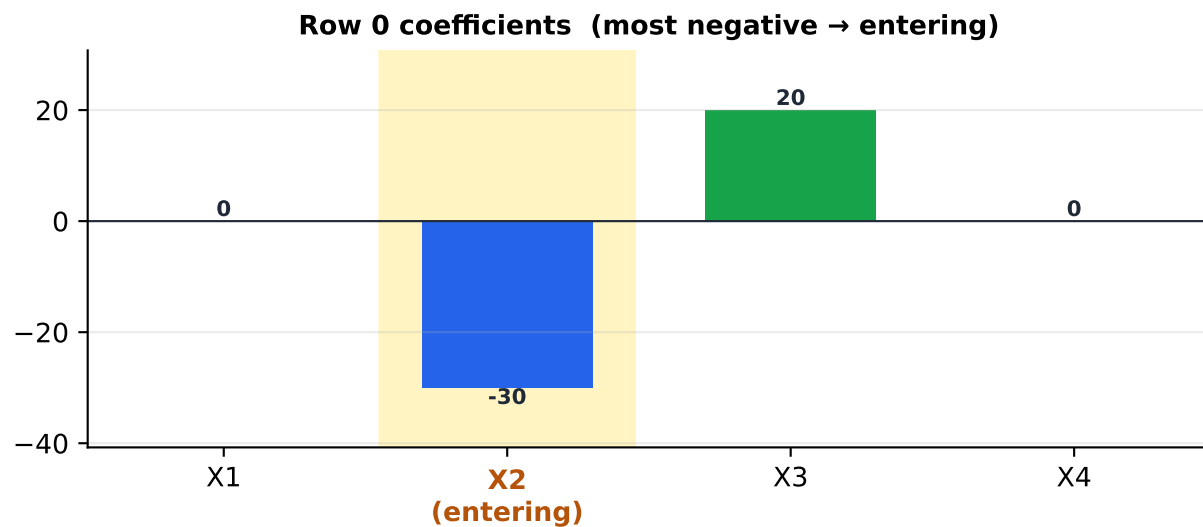
Basic	Eq.	Z	X1	X2	X3	X4	RHS
Z	0	1	0	-30	20	0	20
X1	1	0	1	-1	1	0	1
X4	2	0	0	4	-3	1	4

Row operations:

- Add $20 \times R1$ to R0
- Subtract $3 \times R1$ from R2

Iteration 2: entering & leaving variables

Basic	Eq.	Z	X1	X2	X3	X4	RHS	Ratio
Z	0	1	0	-30	20	0	20	
X1	1	0	1	-1	1	0	1	-ve
X4	2	0	0	4	-3	1	4	1



Iteration 2 - selecting variables.

- Is the current solution optimal? No - row 0 has at least one negative coefficient.
- Entering basic variable: X2 - it has the largest (in absolute value) negative coefficient in row 0 (-30).
- Minimum ratio test (RHS ÷ positive entries of the X2-column): R1: -ve, R2: 1.
- Leaving basic variable: X4 (row 2, smallest ratio). Pivot number = 4.

Iteration 2: pivoting (Gauss-Jordan row operations)

(a) pivot row scaled: Divide R2 by the pivot number 4

Basic	Eq.	Z	X1	X2	X3	X4	RHS	Ratio
Z	0	1	0	-30	20	0	20	
X1	1	0	1	-1	1	0	1	-ve
X4	2	0	0	1	-0.75	0.25	1	1

(b) X2 replaced X4 in the basis. Z = 50

Basic	Eq.	Z	X1	X2	X3	X4	RHS
Z	0	1	0	0	-2.5	7.5	50
X1	1	0	1	0	0.25	0.25	2
X2	2	0	0	1	-0.75	0.25	1

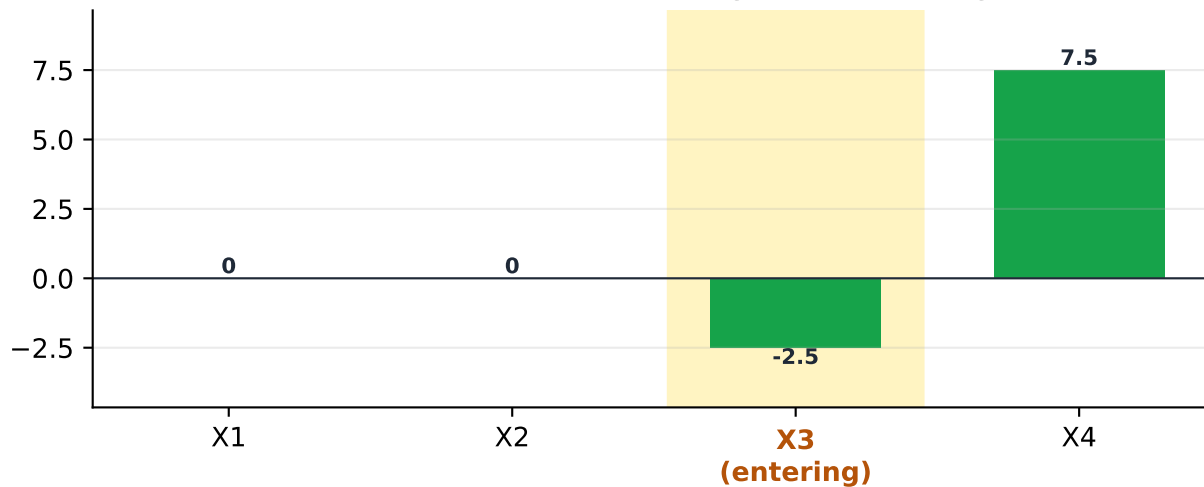
Row operations:

- Add $30 \times R2$ to R0
- Add $1 \times R2$ to R1

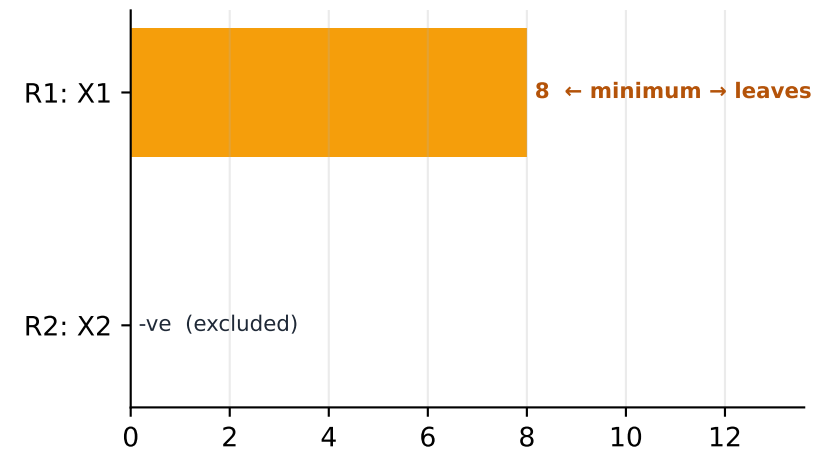
Iteration 3: entering & leaving variables

Basic	Eq.	Z	X1	X2	X3	X4	RHS	Ratio
Z	0	1	0	0	-2.5	7.5	50	
X1	1	0	1	0	0.25	0.25	2	8
X2	2	0	0	1	-0.75	0.25	1	-ve

Row 0 coefficients (most negative → entering)



Minimum ratio test (RHS ÷ X3-column)



Iteration 3 - selecting variables.

- Is the current solution optimal? No - row 0 has at least one negative coefficient.
- Entering basic variable: X3 - it has the largest (in absolute value) negative coefficient in row 0 (-2.5).
- Minimum ratio test (RHS ÷ positive entries of the X3-column): R1: 8, R2: -ve.
- Leaving basic variable: X1 (row 1, smallest ratio). Pivot number = 0.25.

Iteration 3: pivoting (Gauss-Jordan row operations)

(a) pivot row scaled: Divide R1 by the pivot number 0.25

Basic	Eq.	Z	X1	X2	X3	X4	RHS	Ratio
Z	0	1	0	0	-2.5	7.5	50	
X1	1	0	4	0	1	1	8	8
X2	2	0	0	1	-0.75	0.25	1	-ve

(b) X3 replaced X1 in the basis. Z = 70

Basic	Eq.	Z	X1	X2	X3	X4	RHS
Z	0	1	10	0	0	10	70
X3	1	0	4	0	1	1	8
X2	2	0	3	1	0	1	7

Row operations:

- Add $2.5 \times R1$ to R0
- Add $0.75 \times R1$ to R2

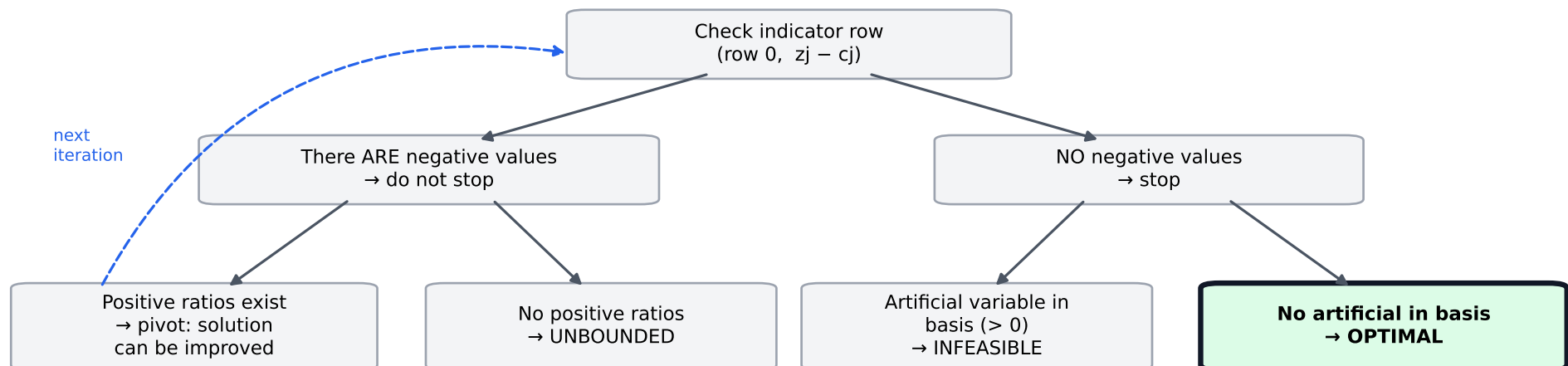
Final tableau - OPTIMAL

Basic	Eq.	Z	X1	X2	X3	X4	RHS
Z	0	1	10	0	0	10	70
X3	1	0	4	0	1	1	8
X2	2	0	3	1	0	1	7

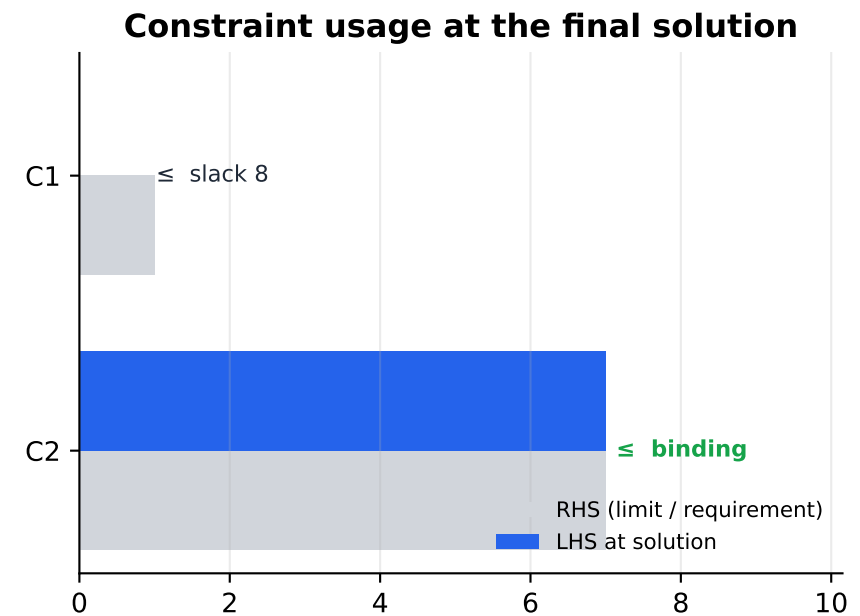
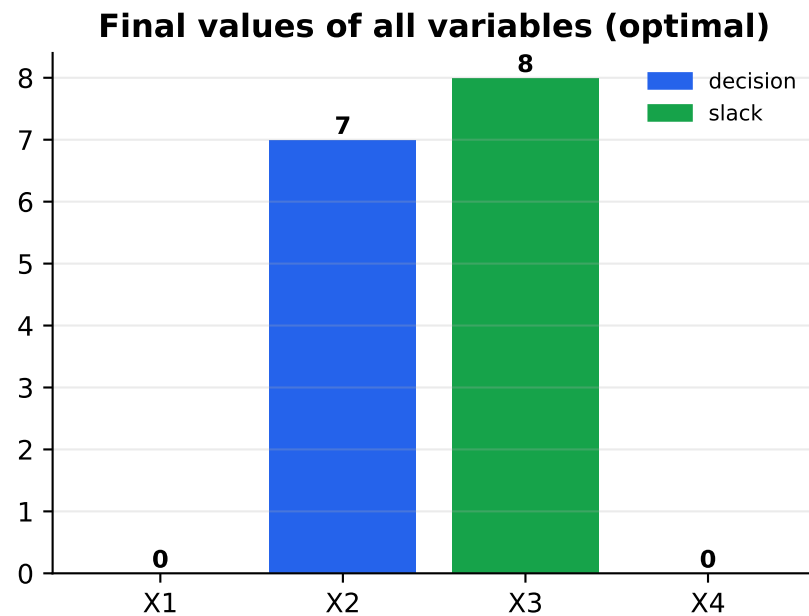
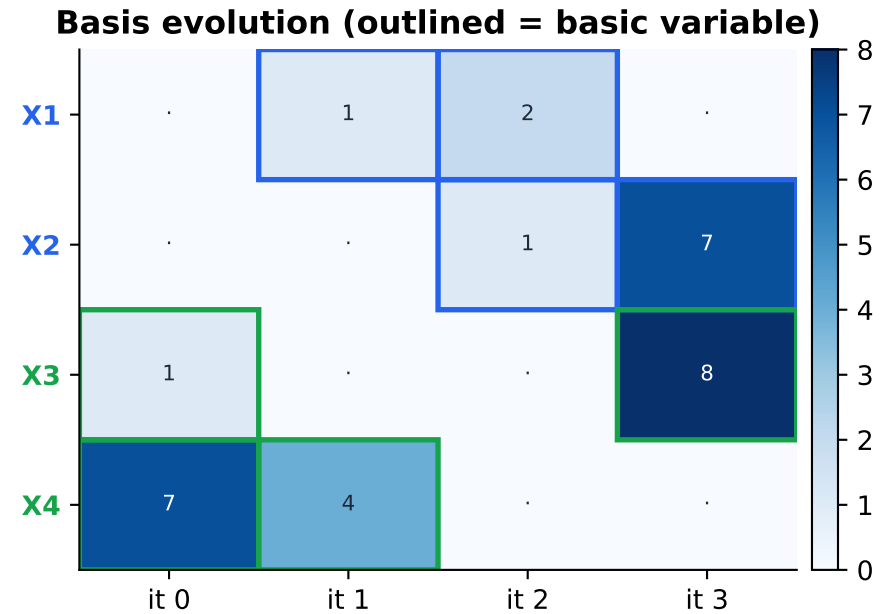
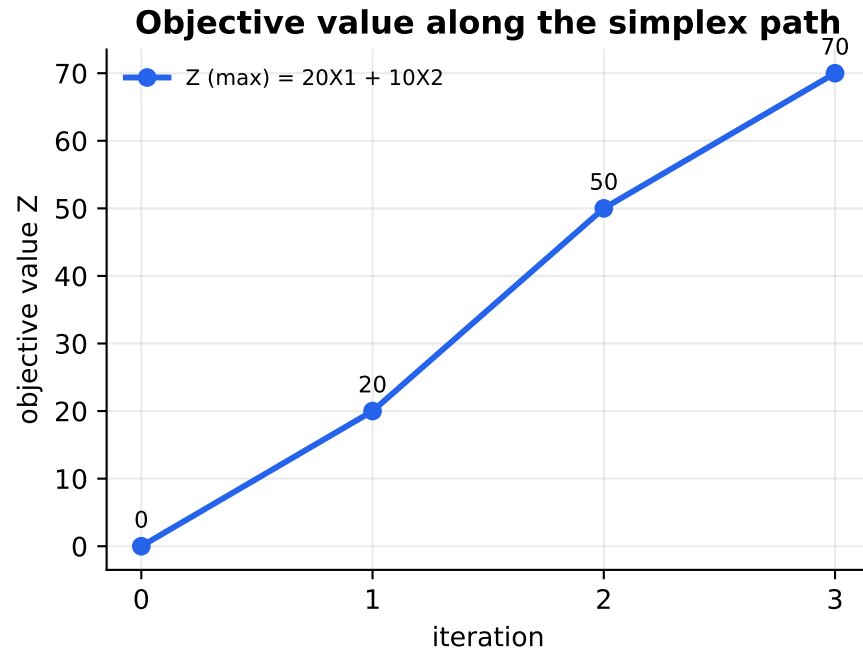
Final check.

- Optimal: row 0 has no negative coefficients. $Z^* = 70$

Simplex decision diagram - outcome: OPTIMAL



Lecture Example 2 - Tabular Simplex - OPTIMAL, $Z^* = 70$



Independent verification with SciPy (HiGHS)

quantity	our tableau simplex	SciPy HiGHS
status	optimal	optimal
Z	70	70
X1	0	0
X2	7	7

✓ Results agree